

Project Implementation Plan for Emotional AI Classroom Insight System (EACIS)

1. Project Overview

This document outlines the development plan for the Emotional AI Classroom Insight System (EACIS). The goal is to design and build a prototype that can analyze student emotions and engagement in real time using video input.

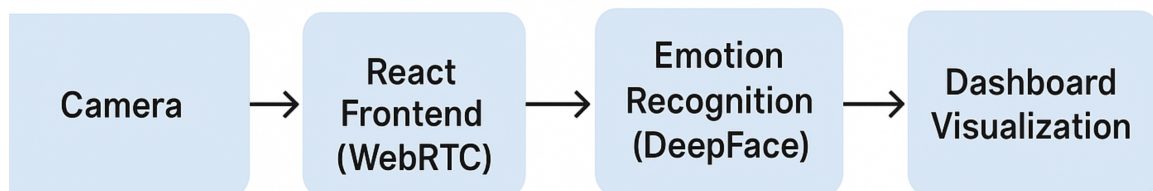
The project will focus on creating an MVP that demonstrates the concept with at least one student in the video.

2. Objectives

- Build a working MVP that detects facial emotions from a live video feed.
- Convert detected emotions into engagement insights.
- ~~Display engagement data through a real-time dashboard.~~
- ~~Ensure privacy and ethical handling of all visual data.~~

3. System Design Plan

The system will be built with separate frontend and backend components. The backend will handle emotion detection and engagement computation, while the frontend will focus on visualization and user interaction.



Frontend Plan

The frontend will be implemented using React with WebRTC for video capture. It will send frames periodically to the backend and receive engagement scores over WebSocket or REST. Charts and indicators will show live emotional and engagement states in a simple dashboard.

Backend Plan

The backend will be developed using FastAPI in Python. It will process incoming frames using OpenCV and run emotion recognition model (DeepFace) to classify emotions. Engagement levels will be computed from these emotions and streamed to the frontend for display.

4. Technologies to Be Used

Component	Planned Technology
Frontend	React, WebRTC, Recharts
Backend	Python, FastAPI, OpenCV
Emotion Model	DeepFace pre-trained CNN
Communication	WebSocket / REST API
Visualization	Recharts / Plotly
Deployment	Local environment or edge device

5. Engagement Computation Plan

A lightweight rule-based approach will be used initially, mapping detected emotions to engagement scores.

Emotion Label	Engagement Score
Happy	1.0
Surprise	0.9
Neutral	0.7
Sad	0.3
Angry	0.2
Fear	0.1

6. Development Roadmap

The implementation will proceed in phases:

Phase 1 (Oct 13 – Oct19)

Research & Planning: Review similar works, identify datasets, finalize architecture.

Phase 2 (Oct 20 – Nov 9)

Backend Setup: Configure FastAPI, integrate emotion recognition models, define API endpoints.

Phase 3 (Nov 10 – Nov 23)

Frontend Setup: Implement webcam capture, live dashboard, and backend connection.

Phase 4 (Nov 24 – Dec 7)

Integrate all modules, test and document findings.

Phase 5 (Dec 8 – Dec 11)

Present MVP